AI technical spike

Farshore: Arcade Trials / October 8, 2026

Model and method

Llama 3.2 3B through local Ollama. Six sequential current-use-case samples using the production adapter from a standalone script. Strict JSON schema, temperature 0, 100 output-token cap, no streaming. Timings include inference, parsing and validation.

Measured latency

Minimum 7.317 s, average 8.279 s, maximum 11.613 s. Individual calls: S1 Cyclops 11.613 s, S2 Circe 7.597 s, S3 Sirens 7.317 s, S4 Strait 7.889 s, S5 Bow 7.614 s, S6 left-side Cyclops 7.641 s. One sequential run, no repeated warm/cold or concurrency benchmark.

Output quality

Six legal planId/focus outputs. Circe after two losses selects chain. Synthetic S6 retains the server-constrained left focus. Current games save zero lane durations. First-time Cyclops and one-loss Strait differ from advisory preferences. Legality does not prove adaptation, fairness or enjoyment. Siren targeting uses current-round misses, starting fresh each round.

Actual failures

E1: nonexistent provider model returns HTTP 404 and model-not-found JSON. E2: malformed provider JSON returns HTTP 400 with unexpected EOF. E3: unsupported difficulty returns application HTTP 400 before provider contact. Exact bodies are archived with sample inputs and raw outputs.

Rate limits and cost

No provider 429 or sustained quota/concurrency threshold was measured. The app maps a provider 429 to a visible busy-service error. These local requests used no billed inference API. Hardware, electricity and future hosting costs remain unmeasured.

Design response and model decision

Show elapsed preparation and Cancel. Run no gameplay clock during preparation. Review before play and offer regeneration, edit/rejection and explicit local fallback. Reject illegal plans before scores change. Retain Llama provisionally for this bounded task, pending player evaluation. No confidence percentages.

Evidence

Exact inputs, outputs and failures: evidence/phase2-arcade-spike-2026-10-08.json. Analytic review: quality-review.json. Code: scripts/phase2-arcade-spike.mjs and server/arcade.ts. Format reference: Ollama structured outputs documentation, accessed October 8.